

Air University



Digital Logic Design (Lab)

Year 2025

Ms. Faryal Naeem

Academic Year: 2024 – 2028

Department of Cyber Security

USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

Date of Submission: April 21, 2025

Experiment No. 08

Title: Multiplexer and Demultiplexer

Objectives

- Gain a solid understanding of the workings of multiplexers and demultiplexers.
- Design and construct circuits for both multiplexers and demultiplexers using logic gates.
- Test their functionality through truth tables and simulations in Proteus.

Equipment and Components

- Proteus Software (for simulations)
- Logic Trainer Kit (for hands-on practice)
- Logic Gate ICs (7408 - AND, 7432 - OR, 7404 - NOT)
- Breadboard
- Power Supply (5V)
- Connecting Wires
- Multimeter (optional)

Introduction

Multiplexers and demultiplexers are crucial circuits in digital systems for managing data flow. A multiplexer (MUX) selects one input from multiple sources and routes it to a single output based on selection signals, which helps to minimize the number of data lines required and simplifies communication. Conversely, a demultiplexer (DEMUX) takes data from one input line and directs it to one of several output lines, also using selection signals, acting as a data distributor. These circuits find applications in areas such as wireless communication, satellite systems, memory addressing, and high-speed data transmission. This experiment focuses on designing, simulating, and testing these circuits to ensure their correct operation through truth tables and Proteus simulations.

Procedure

Multiplexer (MUX):

Design:

- Use two AND gates, one NOT gate, and one OR gate for the circuit.
- Connect input I_1 to the first AND gate along with the NOT output of the select line ($\bar{S}$).
- Link input I_2 to the second AND gate along with the select line (S).
- The outputs from both AND gates are then combined using an OR gate to produce the final output Y .

Implementation:

- Place the necessary integrated circuits (ICs) on the Logic Trainer Kit or within Proteus.
- Properly connect inputs I_1 , I_2 , and the select line S .

Verification:

- Test all possible combinations of inputs and the select line.
- Check the output Y to ensure it matches the truth table accurately.

Demultiplexer (DEMUX):**Design:**

- Utilize two AND gates along with a NOT gate.
- Input Din should be fed into both AND gates.
- Connect the select line S to one AND gate and the NOT output of S ($\bar{S}$) to the other AND gate.
- Outputs Y_0 and Y_1 are produced from each respective AND gate.

Implementation:

- Set up the required ICs on the Logic Trainer Kit or in Proteus.
- Make the connections for the input I and the select line S as specified.

Verification:

- Test all combinations of I and S.
- Review the outputs Y_0 and Y_1 to confirm their accuracy against the truth table.

Logic Expression and Truth Table**Multiplexer (2-to-1) Logic Expression:**

$$Y = \bar{S} \cdot I_1 + S \cdot I_2$$

Truth Table:

S	$\bar{S}$	I_1	I_2	Y
0	1	0	×	0
0	1	1	×	1
1	0	×	0	0
1	0	×	1	1

Note: “ × ” denotes a don't care condition.

Demultiplexer (1-to-2) Logic Expression:

$$Y_0 = \bar{S} \cdot I$$

$$Y_1 = S \cdot I$$

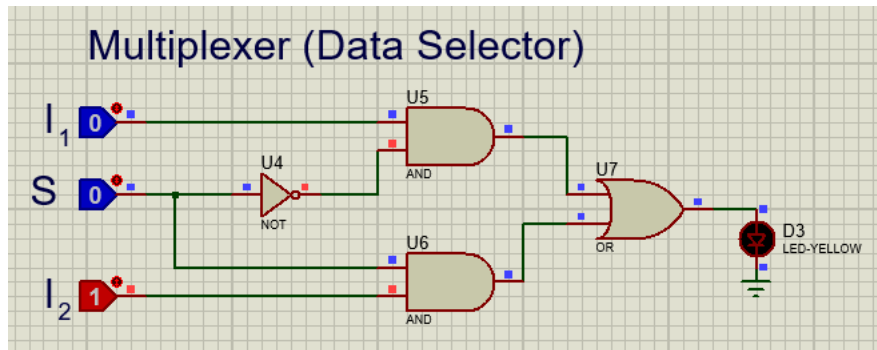
Truth Table:

S	$\bar{S}$	I	Y_0	Y_1
0	1	0	0	0
0	1	1	1	0
1	0	0	0	0
1	0	1	0	1

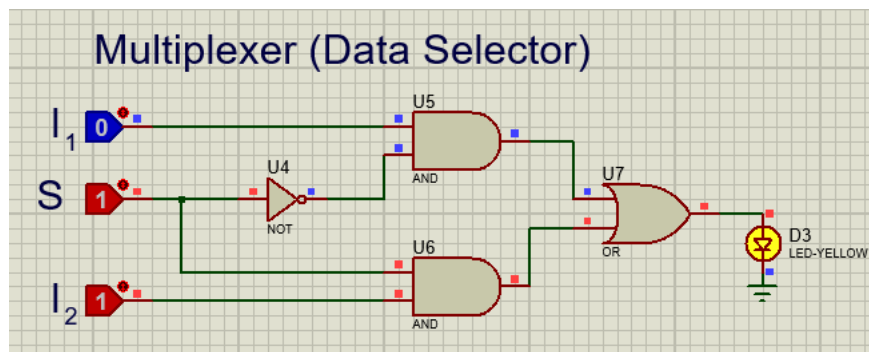
Simulated Results

Multiplexer:

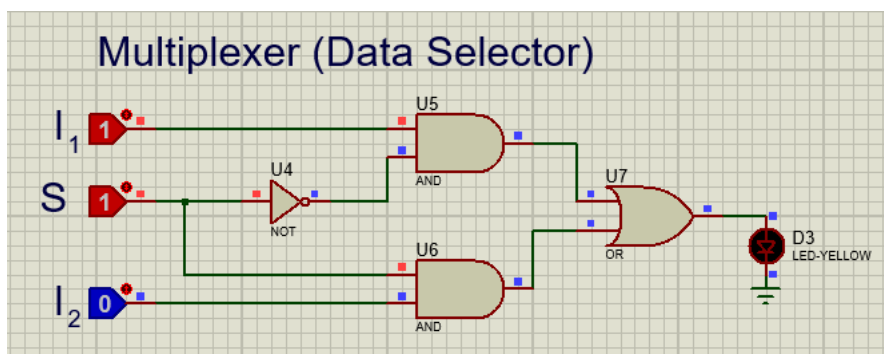
Input: $S = 0$, $I_1 = 0$, $I_2 = \times \rightarrow$ Output: $Y = 0$



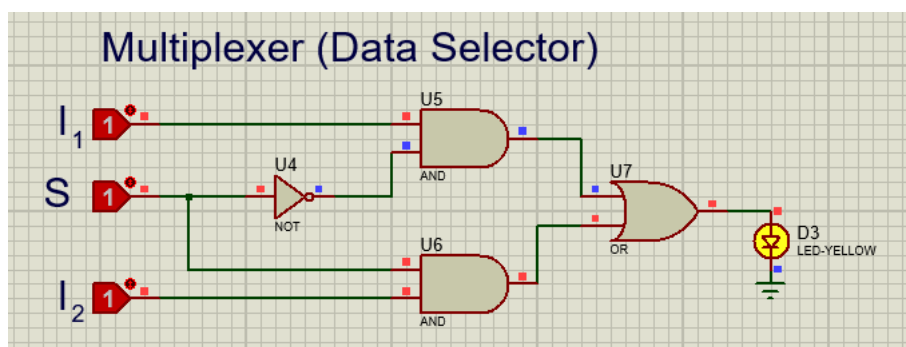
Input: $S = 0$, $I_1 = 1$, $I_2 = \times \rightarrow$ Output: $Y = 1$



Input: $S = 1$, $I_1 = \times$, $I_2 = 0 \rightarrow$ Output: $Y = 0$

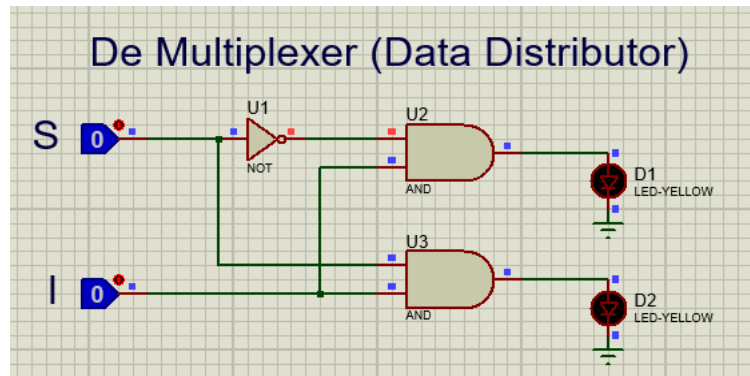


Input: $S = 1$, $I_1 = \times$, $I_2 = 1 \rightarrow$ Output: $Y = 1$

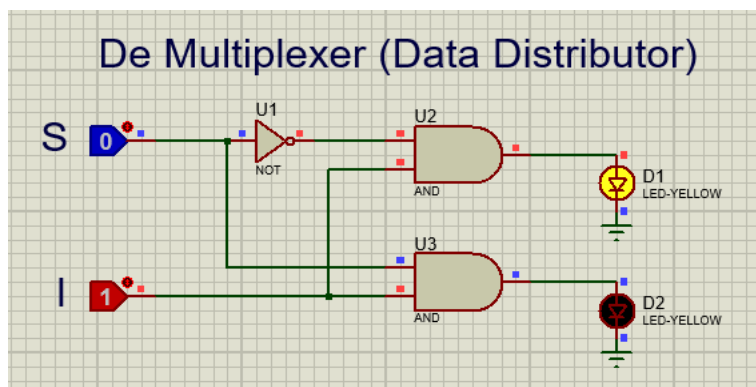


Demultiplexer:

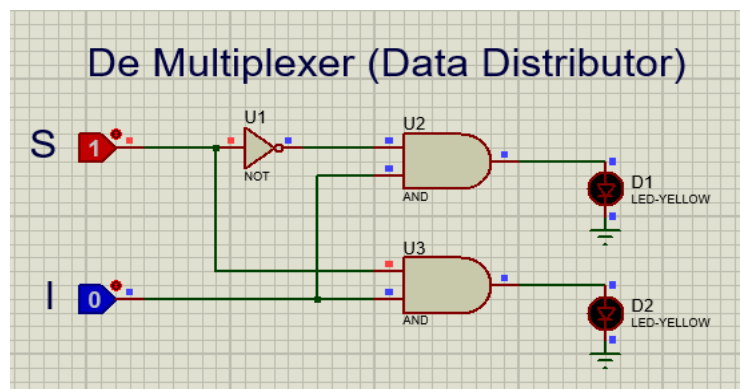
Input: $S = 0, I = 0 \rightarrow$ Output: $Y_0 = 0, Y_1 = 0$



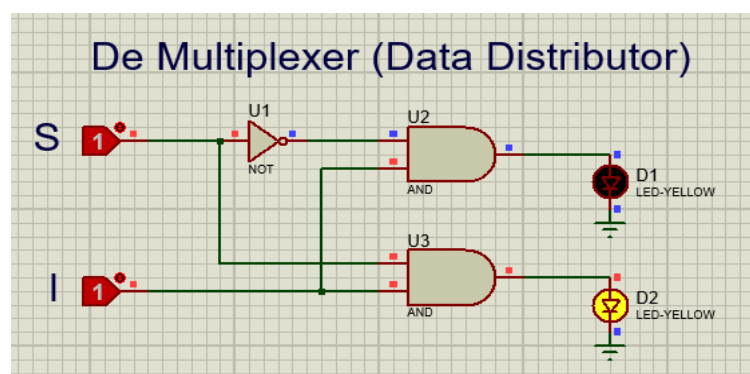
Input: $S = 0, I = 1 \rightarrow$ Output: $Y_0 = 1, Y_1 = 0$



Input: $S = 1, I = 0 \rightarrow$ Output: $Y_0 = 0, Y_1 = 0$



Input: $S = 1, I = 1 \rightarrow$ Output: $Y_0 = 0, Y_1 = 1$



Conclusions

This experiment successfully illustrated how to design, build, and simulate both multiplexer and demultiplexer circuits. The multiplexer efficiently selected one input from several lines based on the selection signals, effectively reducing the communication lines needed. Meanwhile, the demultiplexer accurately directed a single input to one of several outputs using the selection lines. Both circuits are integral to digital communication and data management, simplifying system complexity and enhancing resource utilization. The simulations in Proteus, along with the truth table validations, confirmed the designs' accuracy and reliability.

THE END
